3.4

21. $F(x) = (4x+5)^3 (x^2-2x+5)^4$

$$F'(x) = 3(4x+5)^2 \cdot 4 (x^2-2x+5)^4 + (4x+5)^3 \cdot 4(x^2-2x+5)^3 \cdot (2x-2)$$

$$= 12(4x+5)^2 (x^2-2x+5)^4 + 8(x-1)(4x+5)^3 (x^2-2x+5)^3$$

$$= (x^2-2x+5)^3 (4x+5)^2 [12(x^2-2x+5) + 8(x-1)(4x+5)]$$

22. $G(z) = (1-4z)^2 \sqrt{z^2+1}$

$$G'(z) = 2(1-4z)(-4)\sqrt{z^2+1} + (1-4z)^2 \cdot \frac{1}{2}(z^2+1)^{-\frac{1}{2}} \cdot 2z$$

$$= -8(1-4z)\sqrt{z^2+1} + z(1-4z)^2 \cdot \frac{1}{\sqrt{z^2+1}}$$

$$= \frac{(1-4z)}{\sqrt{z^2+1}} [-8 \cdot (z^2+1) + z(1-4z)]$$

45. $y = e^{\tan\theta}$

$$y' = e^{\tan\theta} \cdot (\tan\theta)'$$

$$= e^{\tan\theta} \cdot \sec^2\theta$$

26. $f(t) = 2^{t^3}$

$$f'(t) = 2^{t^3} \cdot \ln 2 \cdot 3t^2$$

$$= 2 \cdot \ln 2 \, t^2 \cdot 2^{t^3}$$

56. $y = e^{e^x}$

$y' = e^{e^x} \cdot e^x$

$y'' = e^{e^x} \cdot e^x \cdot e^x + e^{e^x} \cdot e^x$

$= e^x \cdot e^{e^x} (e^x + 1)$

60 $y = xe^{-x^2}$, $(0, 0)$

$y' = e^{-x^2} - x \cdot e^{-x^2} \cdot (-2x)$

$= e^{-x^2} + 2x^2 \cdot e^{-x^2}$

$y'|_{x=0} = e^0 + 0 = 1$

$y - 0 = 1(x - 0) \Rightarrow y = x$

#5.5

15. $y\cos x = x^2 + y^2$

$$\frac{dy}{dx}\cos x - y\sin x = 2x + 2y\frac{dy}{dx}$$

$$\frac{dy}{dx}(\cos x - 2y) = 2x + y\sin x$$

$$\frac{dy}{dx} = \frac{2x + y\sin x}{\cos x - 2y}$$

17. $2xe^y + ye^x = 3$

$$2e^y + 2xe^y\cdot\frac{dy}{dx} + \frac{dy}{dx}e^x + y\cdot e^x = 0$$

$$\frac{dy}{dx}(2e^y + 2xe^y + e^x + ye^x) = 0$$

$$\frac{dy}{dx} = 0$$

$$\frac{dy}{dx}(2xe^y + e^x) = -2e^y - ye^x$$

$$\frac{dy}{dx} = \frac{-2e^y - ye^x}{2xe^y + e^x}$$

2(. $e^{x/y} = x - y$

$e^{x/y} \cdot \frac{y - x\frac{dy}{dx}}{y^2} = 1 - \frac{dy}{dx}$

$e^{x/y} \cdot (\frac{1}{y} - \frac{x}{y^2}\frac{dy}{dx}) = 1 - \frac{dy}{dx}$

$e^{x/y} - \frac{x}{y^2}\frac{dy}{dx}e^{x/y} + \frac{dy}{dx} = 1 - e^{x/y}$

$\frac{dy}{dx}(1 - \frac{x}{y^2}e^{x/y}) = 1 - e^{x/y}$

$\frac{dy}{dx} = \frac{1 - e^{x/y}}{1 - \frac{x}{y^2}e^{x/y}}$

22. $g(x) = e^{x^2 \ln x}$

$g'(x) = e^{x^2 \ln x} \cdot (x^2 \ln x)'$

$= e^{x^2 \ln x} \cdot (2x \ln x + x)$

24. $y = \ln \sqrt{\frac{1+2x}{1-2x}}$

$\frac{1}{2} \ln \left(\frac{1+2x}{1-2x}\right)$ 用性质

$\frac{1}{2}[\ln(1+2x) - \ln(1-2x)]$

$\frac{1}{2}\left[\frac{1}{1+2x} \cdot 2 + \frac{2}{1-2x}\right]$

$y' = \frac{1}{\sqrt{\frac{1+2x}{1-2x}}} \cdot \left[\left(\frac{1+2x}{1-2x}\right)^{\frac{1}{2}}\right]'$

$= \left(\frac{1+2x}{1-2x}\right)^{-\frac{1}{2}} \cdot \frac{1}{2}\left(\frac{1+2x}{1-2x}\right)^{-\frac{1}{2}} \cdot \frac{(1-2x)2 - (1+2x)(-2)}{(1-2x)^2}$

$= \frac{1}{2}\left(\frac{1+2x}{1-2x}\right)^{-1} \cdot \frac{4}{(1-2x)^2}$

$= \left(\frac{1+2x}{1-2x}\right)^{-1} \cdot \frac{2}{(1-2x)^2}$

$= \frac{1-2x}{1+2x} \cdot \frac{2}{(1-2x)^2}$

$= \frac{2}{(1+2x)(1-2x)}$